



Small Bodies, Big Impacts: How Climate Change Threatens Children's Health

Abstract

Kids are not little adults. Their bodies are still developing, they breathe faster, and they eat and drink more for their size, so climate change affects them more directly. Research shows children today will face around three times as many climate disasters as their grandparents. This lesson, based on a Harvard Chan C-CHANGE guide, explores how heat, air pollution, disease, and disasters threaten children's health.

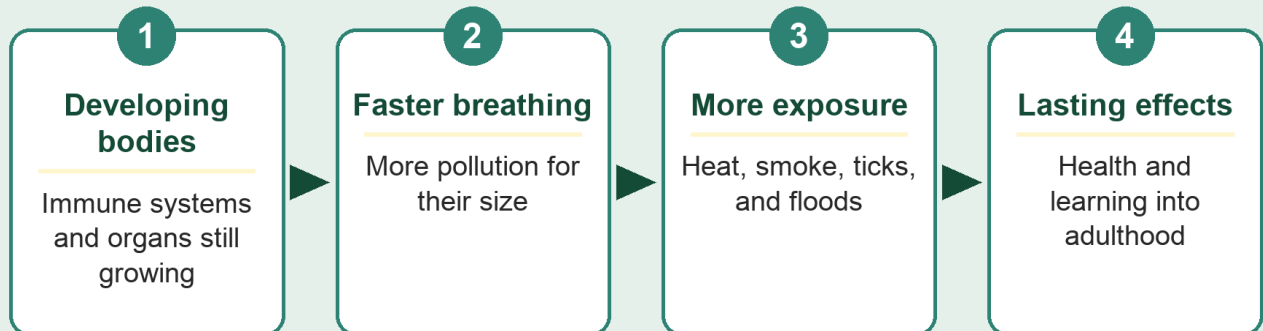
Introduction

Rising greenhouse gases are bringing higher temperatures, more intense storms, and longer droughts, and each of these can deeply affect children. A child's immune system and organs are still developing, and a faster breathing rate means more exposure to harmful air pollutants. Hotter, longer heat waves can make it dangerous to play outside, a serious concern when obesity is already the number one health challenge facing children. Time outdoors can also bring heat stress and more exposure to disease-carrying ticks and mosquitoes. Children also experience climate change emotionally, through trauma from disasters and anxiety about the future. Understanding these risks is the first step toward protecting the next generation.

How Do Scientists Study This?

Scientists combine many types of evidence to understand climate change and children's health. Studies of disease vectors track how ticks and mosquitoes shift their ranges as temperatures rise. Air pollution research estimates how many deaths and illnesses are caused by burning fossil fuels. Mental health studies follow children after storms and fires. Government assessments bring this evidence together with climate projections. A 2023 U.S. Environmental Protection Agency (EPA) report, for example, estimated how children would be affected at 2°C and 4°C of global warming across several areas: asthma, pollen-related emergency visits, Lyme disease, school performance, and displacement from flooding.

Why children face greater climate risks



Track the threats

Scientists monitor disease-carrying insects, air pollution, heat, and extreme weather.

Link to health

Studies connect these threats to asthma, infections, learning, and mental health.

Project the future

Assessments estimate children's health impacts at different levels of warming.

What Did They Discover?

The 2023 EPA report projected that at 2°C to 4°C of warming, childhood asthma cases would rise 4% to 11%, and pollen-related asthma emergency visits 17% to 30% each year. Childhood Lyme disease cases could rise 79% to 241%, or 2,600 to 23,400 new cases per year, as ticks spread north. Heat could reduce academic achievement 4% to 7% each year, and flooding could displace 1 to 2 million or more children. Air pollution already accounts for 20% of newborn deaths worldwide, mostly from low birth weight and preterm birth. Floods can spread diarrheal diseases dangerous to infants, and storms and fires can cause anxiety and post-traumatic stress that raise risks of illness later in life.

Children and climate by the numbers

3×

more climate disasters than their grandparents faced

79–241%

projected rise in childhood Lyme disease cases

20%

of newborn deaths worldwide linked to air pollution

Cutting fossil fuel use can improve air quality quickly, bringing immediate and local benefits to children's health.

Takeaways

Kids are more exposed

Developing bodies and faster breathing make children more vulnerable to climate threats.

Many health risks

Climate change raises risks of asthma, Lyme disease, learning loss, and trauma.

Action helps now

Reducing fossil fuels quickly improves the air children breathe.

Key Terms

Vector-borne disease

An illness spread by insects or ticks, such as Lyme disease, dengue, or malaria.

Lyme disease

A bacterial infection spread by blacklegged ticks, whose range is expanding north as temperatures warm.

Preterm birth

Birth before 37 weeks of pregnancy, which air pollution can make more likely.

Post-traumatic stress

Lasting distress after a frightening event such as a hurricane or wildfire.

Displacement

Being forced to leave home, temporarily or permanently, because of a disaster.

Match-Up: draw a line from each term to its clue

A Vector-borne disease

B Lyme disease

C Preterm birth

D Post-traumatic stress

E Displacement

1 A bacterial infection spread by blacklegged ticks, whose range is expanding north...

2 Being forced to leave home, temporarily or permanently, because of a disaster

3 An illness spread by insects or ticks dengue, or malaria

4 Birth before 37 weeks of pregnancy, which air pollution can make more...

5 Lasting distress after a frightening event such as a hurricane or wildfire

Answer key: A-3 · B-1 · C-4 · D-5 · E-2

Activity

A Child's Climate Health Profile

Students create a profile of how climate change could affect a child at different stages of life and propose protections.

Materials: Profile template with stages: before birth, early childhood, school age, teen; Lesson statistics; Markers

1. For each life stage, list one climate-related health risk from the lesson.
2. Add one statistic from the EPA projections or other findings to support each risk.
3. For each risk, suggest one protection a family, school, or community could provide.
4. Circle the protection that would help the most children at once and explain why.
5. Share profiles and discuss: why do researchers say kids are not little adults?

References

Harvard T.H. Chan School of Public Health, C-CHANGE. Climate, Kids, and Health: Children's Health. <https://hsph.harvard.edu/research/climate-health-c-change/climate-kids-and-health/childrens-health/>